CONTENTS

SL. NO	TOPICS	PAGE NO.
1	ACKNOWLEDGEMENT	1
2	INTRODUCTION OF THE PAGE	2
3	REQUIREMENTS	5
4	PROGRAMMING LANGUAGE AND CONCEPTS USED	6
5	MODULE & FUNCTIONS USED	8
6	DATABASE AND TABLES USED	10
7	PYTHON SOURCE CODE	12
8	MySql DATABASE AND TABLES	33
9	OUTPUT	36
10	REFRENCES	45

ACKNOWLEDMENT

I WANT TO EXPRESS MY DEEPEST APPRECIATION TO THOSE INDIVIDUALS WHO PLAYED CRUCIAL ROLE IN THE SUCCESSFUL COMPLETION OF MY PROJECT

FIRST AND FOREMOST, I WANT TO EXTEND MY SINCERE GRATITUDE TO MY DEVOTE CLASS TEACHER *MRS. TRISHA BHATTACHARYA* WHO CONSTANT SUPPORT AND ENCOURAGE WERE INCREDIBLE VALUABLE THROUGHOUT THIS PROJECT

I ALSO WANT TO THANK PRINCIPAL SR. ROSY FOR CREATING AN ENVIRONMENT THAT POLLUTES ACADEMIC EXPLORATION AND FOR PROVIDING THE NECESSARY RESOURCE

NAME : MOHD FAHAD

INTRODUCTION OF THE PROJECT

THE TOPIC OF THE PROJECT IS **RAILWAY RESERVATION SYSTEM**. THE RAILWAY RESERVATION SYSTEM FACILITATES THE PASSENGERS TO ENQUIRE ABOUT THE TRAINS AVAILABILITY ON THE BASIS OF SOURCE AND DESTINATION, BOOKING AND CANCELLATION OF TICKETS, ENQUIRY ABOUT THE STATUS OF BOOK TICKET, ETC. THE AIM OF CASE STUDY IS TO DESIGN AND DEVELOP A DATABASE MAINTAINING RECORDS OF DIFFERENT TRAINS, TRAINS STATUS AND PASSENGERS. THIS PROJECT

CONTAINS INTRODUCTION TO THE RAILWAY RESERVATION SYSTEM. IT IS THE COMPUTERIZED SYSTEM OR RESERVING THE SEAT OF TRAIN IN ADVANCE. IT IS MAINLY USEFUL FOR A LONG ROUTE. ONLINE RESERVATION HAS MADE THE PROCESS FOR THE RESERVATION OF SEATS VARY EASIER THAN EVER BEFORE.

REQUIREMENTS

➢ HARDWARE REQUIREMENTS:-

- The hardware requirements are laptop/computer
- RAM- 4.00 GB32-bit operating system

➢ SOFTWARE REQUIREMENTS:

- A window based operating system with DOS support
- Python (latest version)
- MySql (latest version)

➢ PROCESSOR REQUIREMENTS:

- Intel(R) Core(TM) i5-4300M CPU @ 2.60GHz

➢ MEMORY REQUIREMENTS:

- Hard disk of about 512GB

PROGRAMMING LANGUAGE AND CONCEPTS USED

The programming language of this project is spied the concepts that we have used in our project are we have taken the use of my SQL connectivity for this project we had made a database in SQL with the help of Python programming and perform every task in SQL by Python programming only we made various types of user defined functions or the functions of each and every functioning in the program we made the use of FLF and else approx. in every function for condition statement we have also imported mysql.connector ,datetime, time, random, pandas, matplotlib.pyplot, os modules.

mysql.connector is use to make a connection between python and MySql, datetime module is used to get current date and time whereas time module is used to get the current time, random module is used to generate random numbers ,pandas are used to arrange the data, mathplotlib.pyplot is used to draw graph.

MODULES & FUNCTIONS USED

MODULE NAME	FUNCTIONS	DESCRIPTION
random	random()	Used to generate captcha
	randint()	Used to generate a Pnr number in a given range
msql.connector	connect()	To make a connection between mysql and python
	cursor()	It provides a cursor object for the working of the database
User Defined Functions	railmenu()	It regulates all the activities like ticket booking, train between stations, train list, Information about book ticket ,ticket cancellation fare of ticket, analytics and exit
	Ticket_booking()	This function is sues to book the ticket of the train
	Available_ALL_Trains()	This function is used to show available trains
	btw_Stations()	This function issues to show trains between two stations
	self_booked_ticket()	This function is used to get the details of the ticket booked
	ticket_cancellation()	This function is used canceled the ticket

	fare_cal()	This function is used to know about the fare
	Analytics()	This function issues to get the analytics of passengers ,fare of trains & average distanced covered by train
	bar_plot()	This function is used to draw a Bar graph of “fare of passengers”
	line_plot()	This function is used to draw a line graph of “AC fare of trains”
	barh_plot()	This function is used to draw a bar graph of “sleeper fare of trains”
	line1_plot()	This function is used to draw a line graph of “average distance covered by train”

DATABASE & TABLES USED

The database used in this project is "railway_reservation_system"

TABLE NAME	FIELD NAME	DATA TYPE	DESCRIPTION
avail_train	Srno	int	Its store serial number
	Train_No	int	It stores the train no
	Train_Name	Varchar(250)	It stores the name of the train
	Train_from	Varchar(200)	It stores the source station of the train
	Train_to	Varchar(300)	it stores the last station of the train
Passenger_details	Pnr_Number	int	It stores the generated pnr number
	Psn_Name	Varchar(25)	Stores the passenger name
	Psn_Ph_no	Varchar(10)	Stores the phone number of the passenger
	Psn_age	int	Stores the age of the passenger
	Psn_gender	Char(1)	Stores the gender of the passenger

	Train_from	Varchar(20)	Stores the source station of the train
	Train_to	Varchar(20)	Stores the destination of the train
	Scheduled_date	date	Stores the date on which the train is scheduled
	Email_ID	Varchar(30)	Source that email id of the passenger
	Fare	int	Stores ticket fair

PYTHON SOURCE CODE

```python
from random import random

import time


while True:

    a1,a=int(input("Enter the Admin Password Of Railway Reservation.:    🔒 ")),2

    while a>1:

        code=random()*10000

        code,code=code//1,int(code)

        print("\n\n==============\ncaptcha🕶=",code,"\n==============")

        captcha=int(input("enter captcha:  🕶 "))


        if captcha==code:

            break

        else:

            print("invalid captcha!!!\ntry again")


    if a1==123:

       for k in range(3):

        print("\t\t\t\t\t\t\t\t\tLogging in.  ",end="\r")        #\r ➡️ carriage return

        time.sleep(0.35)

        print("\t\t\t\t\t\t\t\t\tLogging in.. ",end="\r")

        time.sleep(0.35)

        print("\t\t\t\t\t\t\t\t\tLogging in...",end="\r")

        time.sleep(0.35)

       print(" "*77)

       import os
```

```
49
50      os.system('cls')
51
52
53      try:
54
55              import mysql.connector as msc
56
57              import datetime
58
59              import time
60
61              import random as rd
62
63              import pandas as pd
64
65              import matplotlib.pyplot as plt
66
67              import os
68
69
70              mydb=msc.connect(host="localhost",
71
72                      user="root",
73
74                      password="123456",
75                      )
76
77              crs=mydb.cursor()
78
79              qry01="CREATE DATABASE IF NOT EXISTS railway_reservation_system"
80
81              crs.execute(qry01)
82
83              qry02="USE railway_reservation_system"
84
85              crs.execute(qry02)
86
87
88              qry03="""CREATE TABLE IF NOT EXISTS Passenger_Details (Pnr_Number int(10)
89                primary key,Psn_Name varchar(25),Psn_Ph_no VARCHAR(10),Psn_age int(10),
90                Psn_gender char(1),Train_from varchar(20),Train_to varchar(20),
91                Scheduled_Date DATE,Email_ID varchar(30),Fare int(12))"""
92
93              crs.execute(qry03)
94
95
```

```
qry21="""INSERT INTO Passenger_Details   VALUES   (1021959091,'MOHAN',
'9658256315',25,'M','AJMER','DELHI','2021-03-09','Mohan@gmail.com',720)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021959091"""

qry22="""INSERT INTO Passenger_Details   VALUES   (1021959311,'ROHIT',
9652256315,21,'M','AJMER','AJMER','2021-03-29','Rohit@gmail.com',452)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021959311"""

qry23="""INSERT INTO Passenger_Details   VALUES   (1021959332,'CARL',
6235256363,20,'F','AGRA','MATHURA','2021-05-19','Carl@gmail.com', 1123)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021959332"""

qry24="""INSERT INTO Passenger_Details   VALUES   (1021959301,'MARK',
6582563151,26,'M','SHIMLA','BANGLORE','2021-05-05','Mark11@gmail.com',558)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021959301"""

qry25="""INSERT INTO Passenger_Details   VALUES   (1021951231,'VON',
7584635269,24,'M','CHENNAI','KANYKUMARI','2021-03-25','Von@gmail.com',320)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021951231"""

qry26="""INSERT INTO Passenger_Details   VALUES   (1021959441,'RONIT',
8529639636,21,'M','HALDIA','DELHI','2021-05-09','Ronit45@gmail.com',750)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021959441"""

qry27="""INSERT INTO Passenger_Details   VALUES   (1021952331,'SUNIT',
6589262529,22,'M','LUCKNOW','VARANASI','2021-06-23','Sunit11@gmail.com',950)
 ON DUPLICATE KEY UPDATE Pnr_Number=1021952331"""

crs.execute(qry21)

crs.execute(qry22)

crs.execute(qry23)

crs.execute(qry24)

crs.execute(qry25)

crs.execute(qry26)

crs.execute(qry27)

qry04="USE railway_reservation_system"

crs.execute(qry04)

qryo2="""CREATE TABLE IF NOT EXISTS avail_Train(Srno int(10),Train_No int(15)
primary key,Train_Name varchar(250),Train_from varchar(200),Train_to varchar(300))"""

crs.execute(qryo2)
```

```
qry1="""INSERT INTO avail_Train VALUES  (1001,12847,'Howrah - Digha SUPER AC
  Express','Howrah','Digha') ON DUPLICATE KEY UPDATE Train_No =12847"""

qry2="""INSERT INTO avail_Train  VALUES  (1002,12009,'Mumbai Central - Ahmedabad
  Shatabdi Express','Mumbai','Ahmedabad') ON DUPLICATE KEY UPDATE Train_No =12009"""

qry3="""INSERT INTO avail_Train  VALUES  (1003,120010,'New Delhi - Lucknow Jn Swarn
  Shatabdi Express','New Delhi'  ,'Lucknow ') ON DUPLICATE KEY UPDATE Train_No=120010"""

qry4="""INSERT INTO avail_Train  VALUES  (1004,1280050,'New Delhi - Amritsar Shatabdi
  Express','New Delhi','Amritsar') ON DUPLICATE KEY UPDATE Train_No=1280050"""

qry5="""INSERT INTO avail_Train  VALUES  (1005,1282371,'New Delhi - Daurai (Ajmer
) Shatabdi Express','New Delhi','Ajmer') ON DUPLICATE KEY UPDATE Train_No=1282371"""

qry6="""INSERT INTO avail_Train  VALUES  (1006,1204745,'Howrah - Ranchi Shatabdi
  Express','Digha','Ranchi') ON DUPLICATE KEY UPDATE Train_No=1204745"""

qry7="""INSERT INTO avail_Train VALUES  (1007,1284769,'KSR Bengaluru - MGR Chennai
  Central Shatabdi Expres','Bengaluru','Chennai') ON DUPLICATE KEY UPDATE Train_No=1284769"""

qry8="""INSERT INTO avail_Train VALUES (1008,1201789,'New Delhi -
  Kanpur Central Shatabdi Express','New Delhi','Kanpur') ON DUPLICATE
    KEY UPDATE Train_No=1201789"""

qry9="""INSERT INTO avail_Train  VALUES  (1009,100052,'MGR Chennai Central
  - Coimbatore Shatabdi Express','Chennai','Coimbatore') ON DUPLICATE KEY
  UPDATE Train_No=100052"""

qry10="""INSERT INTO avail_Train  VALUES  (1010,1286565,'New Delhi -
Chandigarh Shatabdi Express','New Delhi','Chandigarh')
ON DUPLICATE KEY UPDATE Train_No=1286565"""

qry11="""INSERT INTO avail_Train VALUES   (1011,101285,'Pune -
Secunderabad Shatabdi Express','Pune ','Secunderabad') ON
DUPLICATE KEY UPDATE Train_No=101285"""

qry12="""INSERT INTO avail_Train  VALUES (1012,100212,'New Delhi -
Firozpur Cantt Shatabdi Express','New Delhi','Firozpur')  ON
  DUPLICATE KEY UPDATE Train_No=100212"""

qry13="""INSERT INTO avail_Train  VALUES (1013,130201,'New Delhi - J
ammu Tawi Rajdhani Express','New Delhi','Jammu ')
ON DUPLICATE KEY UPDATE Train_No=130201"""

qry14="""INSERT INTO avail_Train  VALUES  (1014,111000,'KSR Bengaluru -
Hazrat Nizamuddin Rajdhani Express','Bengaluru','Hazrat Nizamuddin')
  ON DUPLICATE KEY UPDATE Train_No=111000"""
```

```
qry15="""INSERT INTO avail_Train  VALUES  (1015,100005,'M.G.R Chennai Central
  - Hazrat Nizamuddin Rajdhani E','Chennai','Hazrat Nizamuddin')
      ON DUPLICATE KEY UPDATE Train_No=100005"""

qry16="""INSERT INTO avail_Train  VALUES (1016,104040,
'New Delhi - Ranchi Rajdhani Express','New Delhi','Ranchi')
    ON DUPLICATE KEY UPDATE Train_No=104040"""

qry17="""INSERT INTO avail_Train  VALUES (1017,121546,
'MGR Chennai Central - Thiruvananthapuram Central','Chennai','Thiruvananthapuram'
    ON DUPLICATE KEY UPDATE Train_No=121546"""

qry18="""INSERT INTO avail_Train   VALUES( 1018,141524,
'Mumbai LTT - Haridwar AC Express (PT)','Mumbai','Haridwar')
ON DUPLICATE KEY UPDATE Train_No=141524"""

qry19="""INSERT INTO avail_Train  VALUES (1019,152634,
'Bhuj - Mumbai Bandra (T.) AC SuperFast Express','Bhuj ','Mumbai')
  ON DUPLICATE KEY UPDATE Train_No=152634"""

qry20="""INSERT INTO avail_Train  VALUES (1020,136524,
'Lucknow - New Delhi AC Superfast Express','Lucknow','New Delhi')
  ON DUPLICATE KEY UPDATE Train_No=136524"""

qry31="""INSERT INTO avail_Train VALUES (1021,1201546,
'Nagpur - Amritsar AC SF Express','Nagpur','Amritsar')
  ON DUPLICATE KEY UPDATE Train_No=1201546"""

qry32="""INSERT INTO avail_Train  VALUES (1022,126352,
'Arunachal AC SF Express','New Delhi','Itanagar')
    ON DUPLICATE KEY UPDATE Train_No=126352"""

qry33="""INSERT INTO avail_Train  VALUES (1023,104152,
'Darshan Express','Howrah','Digha')
  ON DUPLICATE KEY UPDATE Train_No=104152"""

qry34="""INSERT INTO avail_Train  VALUES (1024,100525,
'Visakhapatnam - Secunderabad AC SF Express','Visakhapatnam','Secunderabad')
  ON DUPLICATE KEY UPDATE Train_No=100525"""

qry35="""INSERT INTO avail_Train  VALUES (1025,100121,
'Patna - Ranchi AC Express','Patna','Ranchi')
    ON DUPLICATE KEY UPDATE Train_No=100121"""

qry36="""INSERT INTO avail_Train  VALUES (1026,114754,
'Barmer - Yesvantpur AC Express','Barmer','Yesvantpur')
    ON DUPLICATE KEY UPDATE Train_No=114754"""
```

```
qry37="""INSERT INTO avail_Train  VALUES  (1027,128474,
'Jammu Tawi - Ahmedabad Express','Jammu Tawi','Ahmedabad')
  ON DUPLICATE KEY UPDATE Train_No=128474"""

qry38="""INSERT INTO avail_Train  VALUES (1028,128654,
'Jammu Tawi - Guwahati Amarnath Express (PT)','Jammu Tawi','Guwahati')
    ON DUPLICATE KEY UPDATE Train_No=128654"""

qry39="""INSERT INTO avail_Train  VALUES (1029,1280417
,'Bandra Terminus - Gorakhpur Avadh Express (PT)','Mumbai','Gorakhpur')
  ON DUPLICATE KEY UPDATE Train_No=1280417"""

qry40="""INSERT INTO avail_Train  VALUES (1030,124557,
'Dibrugarh - Amritsar Weekly Express (PT)','Dibrugarh','Amritsar')
  ON DUPLICATE KEY UPDATE Train_No=124557"""

crs.execute(qry1)

crs.execute(qry2)

crs.execute(qry3)

crs.execute(qry4)

crs.execute(qry5)

crs.execute(qry6)

crs.execute(qry7)

crs.execute(qry8)

crs.execute(qry9)

crs.execute(qry10)

crs.execute(qry11)

crs.execute(qry12)

crs.execute(qry13)

crs.execute(qry14)


crs.execute(qry15)

crs.execute(qry16)

crs.execute(qry17)
```

```
295
296         crs.execute(qry18)
297
298         crs.execute(qry19)
299
300         crs.execute(qry20)
301
302         crs.execute(qry31)
303
304         crs.execute(qry32)
305
306         crs.execute(qry33)
307
308         crs.execute(qry34)
309
310         crs.execute(qry35)
311
312         crs.execute(qry36)
313
314         crs.execute(qry37)
315
316         crs.execute(qry38)
317
318         crs.execute(qry39)
319
320         crs.execute(qry40)
321
322         mydb.commit()
323
324         crs.close()
325
326
327         print("")
328
329         print(r"   ____       __  ____  ___            ___  ___  ___            ___        ____")
330         print(r"     |  |\  | |  \   |   |   | |\  |     |   | |   |   |   |   |   | |   | \    / |   |")
331         print(r"     |  | \ | |   |  |   |___| | \ |     |___| |___|   |   |   |   | |___| \__/  |____")
332         print(r"     |  |  \ | |   |  |   |   | |  \ |     |\   |   |   |   |   | /\ | |   |   |       |")
333         print(r"   _|_ |   \| |__/  _|_  |   | |   \|     | \   |   | _|_  |___ |/  \| |   |   |   |____|")
334         print(r"   ____________________________    ____________________________________________")
335
336
337
338         d=datetime.date.today()
339
340         t=datetime.datetime.now()
341
342         print(" ")
343
344         print(" ")
345
```

```
print("     DATE:-",d.strftime("%A, %d %B %Y"))            #where A=day,d=date,B=month,y=year

print(" ")

print("     TIME:-",t.strftime("%H:%M:%S"))                #where H=hour,M=min,S=sec

print("")

print('')

print("    ")

print("    ==================================================================================================================")

print("                                                    WELCOME TO RAILWAY RESERVATION PORTAL")

print("    ==================================================================================================================")

print("    ")



while True:



    print("""
                    ****************************    DASHBOARD    *********************************

                                    +-------------------------------+
                                    |          1.Passenger          |
                                    +-------------------------------+
                                    +-------------------------------+
                                    |          2.EXIT               |
                                    +-------------------------------+                                    """)

    break

e=input(" ||  **SELECT** || :-")



def railsmenu():

            print("""

                                    +====================================+
                                    |      ****  MAIN MENU  ****         |
                                    +====================================+
                                    |                                    |
                                    |  1. Ticket Booking                 |
                                    |  2. Train_btw_Stations             |
                                    |  3. Train List                     |
                                    |  4. Your Ticket                    |
                                    |  5. Ticket Cancellation            |
                                    |  6. Fare                           |
                                    |  7. Analytics                      |
                                    |  8. Exit                           |
                                    |                                    |
                                    +====================================+
                    """)


            x=int(input("""YOUR OPTION:-""))

            os.system("cls")



            if x==1:

                ticket_booking()
```

```
421
422                                 elif x==2:
423
424                                         print(" ")
425
426                                         print(btw_Stations())
427
428
429
430                                 elif x==3:
431
432                                         print(" ")
433
434                                         print("-----FOLLOWING TRAINS ARE AVAILABLE-----")
435
436                                         print(" ")
437
438                                         print(Available_ALL_Trains())
439
440
441
442                                 elif x==4:
443
444                                         print(" ")
445
446                                         print(self_booked_ticket())
447
448
449
450                                 elif x==5:
451
452                                         print(" ")
453
454                                         print(ticket_cancellation())
455
456
457
458                                 elif x==6:
459
460                                         print(" ")
461
462                                         print(fare_cal())
463
464
465
466                                 elif x==7:
467
468                                         print(" ")
469
```

```
470                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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```

```
533
534                 if Psn_gender==("M") or Psn_gender==("F") or Psn_gender==("N"):
535
536                     break
537
538                 else:
539
540                     print(r"        !~!~!~~ M\F\N only ~~!~!~!~")
541
542
543
544             print()
545
546             print('Journey Details__')
547
548             Train_from=input('       From Station.:')
549
550             Train_to=input('       Destination.:')
551
552             Scheduled_Date=input('Enter Date of Journey (YYYY-MM-DD):')
553
554             Pnr_Number="10"+str(Psn_age)+str(rd.randint(100000,999999))
555
556             Email_ID=input("Enter the Email_ID :")
557
558             Fare=rd.randint(400,1200)
559
560             s1="INSERT INTO  Passenger_Details VALUES(%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)"
561
562             data=(Pnr_Number,Psn_Name,Psn_Ph_no,Psn_age,\
563                 Psn_gender,Train_from,Train_to,Scheduled_Date,Email_ID,Fare)
564
565             crs.execute(s1,(data),)
566
567             print()
568
569             print('  ***TICKET BOOKED SUCCESSFULLY***')
570
571             print("\t","Your Pnr Number is: ", Pnr_Number)
572
573             return("")
574
575
576         def Available_ALL_Trains():
577
578             import mysql.connector as msc
```

```
579
580                 mydb=msc.connect(host='localhost',user='root',passwd='123456',\
581                         database='railway_reservation_system')
582
583                 crs=mydb.cursor()
584
585                 mydb.autocommit=True
586
587                 qry="Select * from avail_train "
588
589                 crs.execute(qry)
590
591                 data1 = crs.fetchall()
592
593                 df = pd.DataFrame(data1, columns=["Srno","Train_No",\
594                                 "Train_Name","Train_from","Train_to",])
595
596                 print(df)
597
598                 mydb.commit()
599
600                 crs.close()
601
602                 return("")
603
604
605             def btw_Stations():
606
607                 import mysql.connector as msc
608
609                 mydb=msc.connect(host='localhost',user='root',passwd='123456',\
610                         database='railway_reservation_system')
611
612                 crs=mydb.cursor()
613
614                 mydb.autocommit=True
615
616                 crs.execute("use railway_reservation_system")
617
618
619
620                 Sr_train=input("Enter the Source Station.:")
621
622                 De_train=input("Enter the Destination Station.:")
623
624                 crs.execute('select * from avail_train where Train_from=(%s) \
625                         and Train_to=(%s)',(Sr_train,De_train,))
626
627                 data=crs.fetchall()
```

```
628
629                 a=[]
630
631                 for i in data:
632
633                     a.append(i)
634
635
636
637                 if len(a)!=1:
638
639                     print('~!~!~!~!~~NO DATA FOUND~~!~!~!~!~')
640
641
642                 else:
643
644                     mydb.commit()
645
646                     df = pd.DataFrame(data, columns=["Srno","Train_No","Train_Name",\
647                                                      "Train_from" ,  "Train_to" ])
648
649                     print("")
650
651                     print("")
652
653                     print(df)
654
655                     mydb.commit()
656
657                     crs.close()
658
659                 return("")
660
661
662
663             def self_booked_ticket():
664
665                 import mysql.connector as msc
666
667                 mydb=msc.connect(host='localhost',user='root',passwd='123456',\
668                                  database='railway_reservation_system')
669
670                 crs=mydb.cursor()
671
672                 mydb.autocommit=True
673
674                 Self_Pnr=int(input('Enter your Pnr_number No:'))
675
676                 crs.execute('select * from passenger_details where Pnr_Number=(%s)',(Self_Pnr,)
```

```
677
678                 data=crs.fetchall()
679
680                 a=[]
681
682                 for i in data:
683
684                     a.append(i)
685
686
687
688                 if len(a)!=1:
689
690                     print('~!~!~!~!~~NO DATA FOUND~~!~!~!~!~')
691
692
693
694                 else:
695
696                     mydb.commit()
697
698                     df = pd.DataFrame(data, columns=\
699                                       ["Pnr_Number","Psn_Name","Psn_Ph_no"\
700                                        , "Psn_age" ,  "Psn_gender","Train_from",\
701                                         " Train_to","Scheduled_Date","Email_ID","Fare" ])
702
703                     print("")
704
705                     print("")
706
707                     print(df)
708
709                     mydb.commit()
710
711                     crs.close()
712
713                 return("")
714
715
716
717             def ticket_cancellation():
718
719                 import mysql.connector as msc
720
721                 mydb=msc.connect(host='localhost',user='root',passwd='123456',\
722                                  database='railway_reservation_system')
723
724                 crs=mydb.cursor()
725
```

```
726                 mydb.autocommit=True
727
728                 crs.execute("use railway_reservation_system")
729
730                 Self_Pnr=int(input("Enter Your Pnr_Number.:"))
731
732                 Self_Name=input("Enter Your Name.:")
733
734                 Self_Phno=int(input("Enter You Phone no.:"))
735
736                 qry='DELETE FROM passenger_details WHERE Pnr_Number=(%s) or Psn_Name=(%s) or Psn_Ph_no=(%s)'
737
738
739
740                 s=input("ARE YOU SURE  YOU WANT TO CANCEL THE TICKET (YES/NO): ")
741
742
743
744                 if s in ["YES","yes","Yes"]:
745
746                     crs.execute('DELETE FROM passenger_details WHERE Pnr_Number=(%s) \
747                                 or Psn_Name=(%s) or Psn_Ph_no=(%s)',(Self_Pnr,Self_Name,Self_Phno,))
748
749                     print("")
750
751                     print("        Your Ticket is Cancelled")
752
753
754
755                 else:
756
757                     print("         Ticket NOT Cancelled")
758
759
760
761                 return("")
762
763
764
765             def fare_cal():
766
767                 Sr_train=input("Enter the Source Station.:")
768
769
770                 De_train=input("Enter the Destination Station.:")
771
772                 Fare=rd.randint(400,1200)
773
774                 print("            +==========================================================+")
775
```

urce Control (Ctrl+Shift+G)

```
776                 print("            |        ****  FARE FROM",Sr_train,"TO",De_train,"   ****    ")
779                 print("            +==========================================================+")
780                 print("            |                        ",Fare,"                          ")
781
782                 print("            +==========================================================+")
783
784                 return("")
785
786
787
788
789             def Analytics():
790
791                 print("""
792                         +=====================================================+
793                         |                 ****  MAIN MENU  ****               |
794                         +=====================================================+
795                         |                                                     |
796                         |  1. BAR PLOT :-   Fare  of  Passengers              |
797                         |  2. LINE PLOT :-  AC Fare  Of  Trains               |
798                         |  3. BAR PLOT :-   Sleeper Fare Of  Trains           |
799                         |  4. LINE PLOT :-  Average Distance Covered By Train |
800                         |                                                     |
801                         +=====================================================+
802                     """)
803
804                 opt=int(input("""       YOUR OPTION:-"""))
805
806                 if opt==1:
807
808                         print(" ")
809
810                         print(bar_plot())
811
812
813
814                 elif opt==2:
815                         print(" ")
816
817                         print(line_plot())
818
819
820
```

```
821                 elif opt==3:
822
823                         print(" ")
824
825                         print(barh_plot())
826
827
828
829                 elif opt==4:
830
831                         print(" ")
832
833                         print(line1_plot())
834
835
836
837                 else:
838
839                         print(" ")
840
841                         print("~!~!~!~WRONG CHOICE PLEASE ENTER A VALID VALUE~!~!~!~")
842
843
844
845
846         def  bar_plot():
847
848                 plt.figure(figsize=(9,6))
849
850                 plt.bar(x=["MOHAN","ROHIT","CARL","MARK","VON","RONIT","SUNIT"],
851
852                 height=[720,452,1123,558,320,750,950],color='deeppink')
853
854                 plt.xticks(rotation=45)
855
856                 plt.title("Fare of Passengers")
857
858                 plt.savefig("Fare of Passengers.png")
859
860                 plt.show()
861
862                 return("")
863
864
865
```

```
def line_plot():

        print("Line Plot")

        x=["Goa Express","JammuTavi Superfast","Punjab Mail Express","Dakshin Express","Delhi-Mumbai Shatabdi"

        y=[4500,8500,3600,7600,12000]

        plt.title("AC Fare Of Trains",fontsize=14,color="g")

        plt.xlabel("Trains",fontsize=14,color="g")

        plt.ylabel("AC Fare",fontsize=14,color="g")

        plt.xticks(rotation=12)            #rotate elements of x axis by x°



        plt.plot(x,y,ls="dashed",linewidth=3, color="magenta")

        plt.savefig("AC Fare For Trains.png")

        plt.show()

        return("")



def barh_plot():

        print("Horizontal Bar Plot")

        x=["Goa Express","JammuTavi Superfast","Punjab Mail Express","Dakshin Express","Delhi-Mumbai Shatabdi"

        y=[1500,4500,2600,7600,9000]

        plt.title("Sleeper Fare  Of  Trains",fontsize=14,color="b")

        plt.xlabel("Trains",fontsize=14,color="c")

        plt.ylabel("Sleeper Fare",fontsize=14,color="c")

        plt.barh(x,y, color="pink", edgecolor="black")      #barh is used to draw horizontal bar

        plt.savefig("Sleeper Fare Of Trains.png")

        plt.show()

        return("")
```

```python
def line1_plot():

        print("Line Plot")

        x=["Goa Express","JammuTavi Superfast","Punjab Mail Express","Dakshin Express","Delhi-Mumbai Shatabdi"]

        y=[500,1200,650,750,1350]

        plt.title("Average Distance Covered By Train",fontsize=14,color="c")

        plt.xlabel("Trains",fontsize=14,color="c")

        plt.ylabel("(Distance in Kms)",fontsize=14,color="r")

        plt.plot(x,y, marker="o", label="Avg Distance")      #marker is use to highlight each point on a graph

        plt.xticks(rotation=17)

        plt.savefig("Avearage Distance By Trains.png")

        plt.show()

        return("")



if e=="1" :

    print("\n"*3)

    print("                                                  0%  ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  100%",end="\r")

    time.sleep(1.5)

    print("                                                  15% ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  100%",end="\r")

    time.sleep(1)

    print("                                                  50% ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  100%",end="\r")

    time.sleep(0.75)

    print("                                                  90% ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  100%",end="\r")

    time.sleep(1.5)

    print("                                                  100% ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  100%",end="\r")

    time.sleep(0.5)

    print(" "*90)


    print(railsmenu())


elif e=="2":

        print(" ")

        print("\n"
              "      \n"
              "          ##=================================================##\n"
              "          ||                    THANK YOU                    ||\n"
              "          ##=================================================##\n"
              "\n")
        print('''

DEVELOPED BY MOHD.FAHAD  XII-A 2024-25   (CANOSSA SCHOOL,LUCKNOW)
        ''')
```

```
992
993
994
995                 else:
996
997                     print(" ")
998
999                     print("~!~!~!~PLEASE ENTER 1 OR 2~!~!~!~")
1000
1001                mydb.close()
1002
1003        except():
1004
1005                pass
1006
1007    else:
1008
1009            print("Password is incorrect,\nPlease enter the correct password.")
1010
1011            a2=str(input("Enter any number to again apply for password or press any alphabet to exit:"))
1012
1013            print()
1014
1015            if a2.isalpha():
1016
1017                print("Thanks for using our service.")
1018
1019                break
```

MySql DATABASE AND TABLES

DATABASE : RAILWAY_RESERVATION_SYSTEM

```
Enter password: ******
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.0.36 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use railway_reservation_system
Database changed
mysql>
mysql>
mysql> show tables;
+-------------------------------------+
| Tables_in_railway_reservation_system |
+-------------------------------------+
| avail_train                         |
| passenger_details                   |
+-------------------------------------+
2 rows in set (0.00 sec)

mysql> |
```

TABLE : PASSENGER_DETAILS

```
mysql> select* from passenger_details;
+------------+----------+------------+---------+------------+------------+------------+---------------+-------------------+------+
| Pnr_Number | Psn_Name | Psn_Ph_no  | Psn_age | Psn_gender | Train_from | Train_to   | Scheduled_Date | Email_ID          | Fare |
+------------+----------+------------+---------+------------+------------+------------+---------------+-------------------+------+
| 1021951231 | VON      | 7584635269 |      24 | M          | CHENNAI    | KANYKUMARI | 2021-03-25    | Von@gmail.com     |  320 |
| 1021952331 | SUNIT    | 6589262529 |      22 | M          | LUCKNOW    | VARANASI   | 2021-06-23    | Sunit11@gmail.com |  950 |
| 1021959091 | MOHAN    | 9658256315 |      25 | M          | AJMER      | DELHI      | 2021-03-09    | Mohan@gmail.com   |  720 |
| 1021959301 | MARK     | 6582563151 |      26 | M          | SHIMLA     | BANGLORE   | 2021-05-05    | Mark11@gmail.com  |  558 |
| 1021959311 | ROHIT    | 9652256315 |      21 | M          | AJMER      | AJMER      | 2021-03-29    | Rohit@gmail.com   |  452 |
| 1021959332 | CARL     | 6235256363 |      20 | F          | AGRA       | MATHURA    | 2021-05-19    | Carl@gmail.com    | 1123 |
| 1021959441 | RONIT    | 8529639636 |      21 | M          | HALDIA     | DELHI      | 2021-05-09    | Ronit45@gmail.com |  750 |
+------------+----------+------------+---------+------------+------------+------------+---------------+-------------------+------+
7 rows in set (0.00 sec)
```

TABLE : AVAIL_TRAIN

```
mysql> select* from avail_train;
```

Srno	Train_No	Train_Name	Train_from	Train_to
1002	12009	Mumbai Central - Ahmedabad Shatabdi Express	Mumbai	Ahmedabad
1001	12847	Howrah - Digha SUPER AC Express	Howrah	Digha
1015	100005	M.G.R Chennai Central - Hazrat Nizamuddin Rajdhani E	Chennai	Hazrat Nizamuddin
1009	100052	MGR Chennai Central - Coimbatore Shatabdi Express	Chennai	Coimbatore
1025	100121	Patna - Ranchi AC Express	Patna	Ranchi
1012	100212	New Delhi - Firozpur Cantt Shatabdi Express	New Delhi	Firozpur
1024	100525	Visakhapatnam - Secunderabad AC SF Express	Visakhapatnam	Secunderabad
1011	101285	Pune - Secunderabad Shatabdi Express	Pune	Secunderabad
1016	104040	New Delhi - Ranchi Rajdhani Express	New Delhi	Ranchi
1023	104152	Darshan Express	Howrah	Digha
1014	111000	KSR Bengaluru - Hazrat Nizamuddin Rajdhani Express	Bengaluru	Hazrat Nizamuddin
1026	114754	Barmer - Yesvantpur AC Express	Barmer	Yesvantpur
1003	120010	New Delhi - Lucknow Jn Swarn Shatabdi Express	New Delhi	Lucknow
1017	121546	MGR Chennai Central - Thiruvananthapuram Central	Chennai	Thiruvananthapuram
1030	124557	Dibrugarh - Amritsar Weekly Express (PT)	Dibrugarh	Amritsar
1022	126352	Arunachal AC SF Express	New Delhi	Itanagar
1027	128474	Jammu Tawi - Ahmedabad Express	Jammu Tawi	Ahmedabad
1028	128654	Jammu Tawi - Guwahati Amarnath Express (PT)	Jammu Tawi	Guwahati
1013	130201	New Delhi - Jammu Tawi Rajdhani Express	New Delhi	Jammu
1020	136524	Lucknow - New Delhi AC Superfast Express	Lucknow	New Delhi
1018	141524	Mumbai LTT - Haridwar AC Express (PT)	Mumbai	Haridwar
1019	152634	Bhuj - Mumbai Bandra (T.) AC SuperFast Express	Bhuj	Mumbai
1021	1201546	Nagpur - Amritsar AC SF Express	Nagpur	Amritsar
1008	1201789	New Delhi - Kanpur Central Shatabdi Express	New Delhi	Kanpur
1006	1204745	Howrah - Ranchi Shatabdi Express	Digha	Ranchi
1004	1280050	New Delhi - Amritsar Shatabdi Express	New Delhi	Amritsar
1029	1280417	Bandra Terminus - Gorakhpur Avadh Express (PT)	Mumbai	Gorakhpur
1005	1282371	New Delhi - Daurai (Ajmer) Shatabdi Express	New Delhi	Ajmer
1007	1284769	KSR Bengaluru - MGR Chennai Central Shatabdi Expres	Bengaluru	Chennai
1010	1286565	New Delhi - Chandigarh Shatabdi Express	New Delhi	Chandigarh

30 rows in set (0.00 sec)

OUTPUT

Starting of program:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\LENOVO\Desktop\Java Tutorials> & "C:/Program Files/Python312/python.exe" "c:/Users/LENOVO/Desktop/Java Tutorials/f.py"
Enter the Admin Password Of Railway Reservation.:   🔒 123


==============
captcha 🚔= 7949
==============
enter captcha:   🚔 7949
                                                    Logging in...
```

Home screen:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

 ____   ___   ____   ____        ____  ____  ____      ____   ____
 |  |\  | |  \  |  |  |  |\  |   |  | |  |  |  |  |  |  |  | \    / |  |
 |  | \ | |  |  |  |  |___| | \ |  |___| |___|  |  |  |  |  |___|  \__/  |____
 |  |  \ | |  |  |  |  |  | | \ |  |\  |  |  |  |  |  | /\ |  |  |  |  |
 __|__ |  \| |___/  __|__ |  | |  \|  | \ |  |  |  __|__ |____ |/  \| |  |  |  |  |____|
 ----------------------------------------    ----------------------------------------------------

  DATE:- Thursday, 14 November 2024

  TIME:- 19:43:15


 ==================================================================================================================
                                        WELCOME TO RAILWAY RESERVATION PORTAL
 ==================================================================================================================

                          ****************************    DASHBOARD    ******************************

                                          +-------------------------------+
                                          |          1.Passenger          |
                                          +-------------------------------+
                                          +-------------------------------+
                                          |          2.EXIT               |
                                          +-------------------------------+
 ||  **SELECT** || :-1


                                          90%  ▒▒▒
```

Main menu:

```
+=================================+
|      ****  MAIN MENU  ****      |
+=================================+
|                                 |
|  1. Ticket Booking              |
|  2. Train_btw_Stations          |
|  3. Train List                  |
|  4. Your Ticket                 |
|  5. Ticket Cancellation         |
|  6. Fare                        |
|  7. Analytics                   |
|  8. Exit                        |
|                                 |
+=================================+

YOUR OPTION:-1
```

Ticket Booking :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter  Name.:Mohd Fahad
Enter  Phone No.:9123456789
Enter age.:16

               M=MALE    F=FEMALE        N=NOT TO MENTION
Enter  Gender M/F/N:m

Journey Details__
       From Station.:Lucknow
       Destination.:New Delhi
Enter Date of Journey (YYYY-MM-DD):2024-11-16
Enter the Email_ID :fhd123@gmail.com

  ***TICKET BOOKED SUCCESSFULLY***
         Your Pnr Number is:  1016647326
```

Train Between Station :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter the Source Station.:New Delhi
Enter the Destination Station.:Chandigarh

   Srno  Train_No                          Train_Name Train_from    Train_to
0  1010   1286565  New Delhi - Chandigarh Shatabdi Express  New Delhi  Chandigarh
```

Train List :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

-----FOLLOWING TRAINS ARE AVAILABLE-----

    Srno  Train_No                                  Train_Name     Train_from           Train_to
0   1002     12009        Mumbai Central - Ahmedabad Shatabdi Express      Mumbai          Ahmedabad
1   1001     12847                   Howrah - Digha SUPER AC Express      Howrah              Digha
2   1015    100005  M.G.R Chennai Central - Hazrat Nizamuddin Rajd...     Chennai  Hazrat Nizamuddin
3   1009    100052  MGR Chennai Central - Coimbatore Shatabdi Express     Chennai        Coimbatore
4   1025    100121                        Patna - Ranchi AC Express       Patna            Ranchi
5   1012    100212        New Delhi - Firozpur Cantt Shatabdi Express   New Delhi          Firozpur
6   1024    100525         Visakhapatnam - Secunderabad AC SF Express  Visakhapatnam      Secunderabad
7   1011    101285               Pune - Secunderabad Shatabdi Express       Pune        Secunderabad
8   1016    104040                New Delhi - Ranchi Rajdhani Express   New Delhi            Ranchi
9   1023    104152                                   Darshan Express      Howrah              Digha
10  1014    111000  KSR Bengaluru - Hazrat Nizamuddin Rajdhani Exp...   Bengaluru  Hazrat Nizamuddin
11  1026    114754                    Barmer - Yesvantpur AC Express      Barmer        Yesvantpur
12  1003    120010      New Delhi - Lucknow Jn Swarn Shatabdi Express   New Delhi          Lucknow
13  1017    121546   MGR Chennai Central - Thiruvananthapuram Central     Chennai  Thiruvananthapuram
14  1030    124557           Dibrugarh - Amritsar Weekly Express (PT)   Dibrugarh          Amritsar
15  1022    126352                          Arunachal AC SF Express   New Delhi          Itanagar
16  1027    128474                    Jammu Tawi - Ahmedabad Express   Jammu Tawi        Ahmedabad
17  1028    128654        Jammu Tawi - Guwahati Amarnath Express (PT)  Jammu Tawi         Guwahati
18  1013    130201            New Delhi - Jammu Tawi Rajdhani Express   New Delhi            Jammu
19  1020    136524           Lucknow - New Delhi AC Superfast Express     Lucknow        New Delhi
20  1018    141524              Mumbai LTT - Haridwar AC Express (PT)      Mumbai         Haridwar
21  1019    152634     Bhuj - Mumbai Bandra (T.) AC SuperFast Express       Bhuj            Mumbai
22  1021   1201546                   Nagpur - Amritsar AC SF Express      Nagpur          Amritsar
23  1008   1201789        New Delhi - Kanpur Central Shatabdi Express   New Delhi            Kanpur
24  1006   1204745                  Howrah - Ranchi Shatabdi Express       Digha            Ranchi
25  1004   1280050              New Delhi - Amritsar Shatabdi Express   New Delhi          Amritsar
26  1029   1280417     Bandra Terminus - Gorakhpur Avadh Express (PT)      Mumbai        Gorakhpur
27  1005   1282371        New Delhi - Daurai (Ajmer) Shatabdi Express   New Delhi             Ajmer
28  1007   1284769  KSR Bengaluru - MGR Chennai Central Shatabdi E...   Bengaluru           Chennai
29  1010   1286565            New Delhi - Chandigarh Shatabdi Express   New Delhi         Chandigarh
```

Your Ticket:

```
Explorer (Ctrl+Shift+E)    DEBUG CONSOLE    TERMINAL    PORTS

Enter your Pnr_number No:1016647326


   Pnr_Number    Psn_Name   Psn_Ph_no  Psn_age Psn_gender Train_from   Train_to Scheduled_Date         Email_ID  Fare
0  1016647326  Mohd Fahad  9123456789       16         M    Lucknow  New Delhi     2024-11-16  fhd123@gmail.com   909
```

Ticket Cancellation:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter Your Pnr_Number.:1021959332
Enter Your Name.:CARL
Enter You Phone no.:623525636
ARE YOU SURE  YOU WANT TO CANCEL THE TICKET (YES/NO): YES

        Your Ticket is Cancelled
```

FAIR :

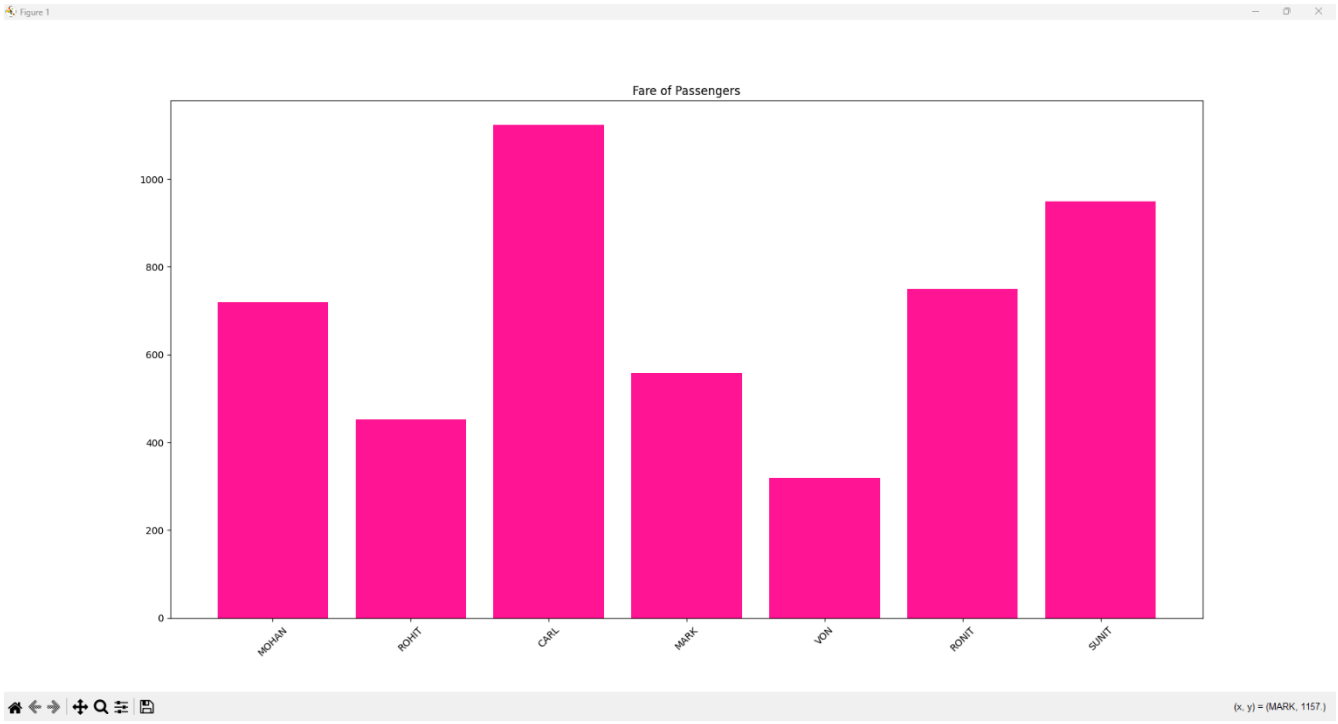
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter the Source Station.:Howrah
Enter the Destination Station.:Lucknow
            +==============================================================+
            |        ****  FARE FROM Howrah TO Lucknow    ****
            +==============================================================+
            |                           778
            +==============================================================+
```

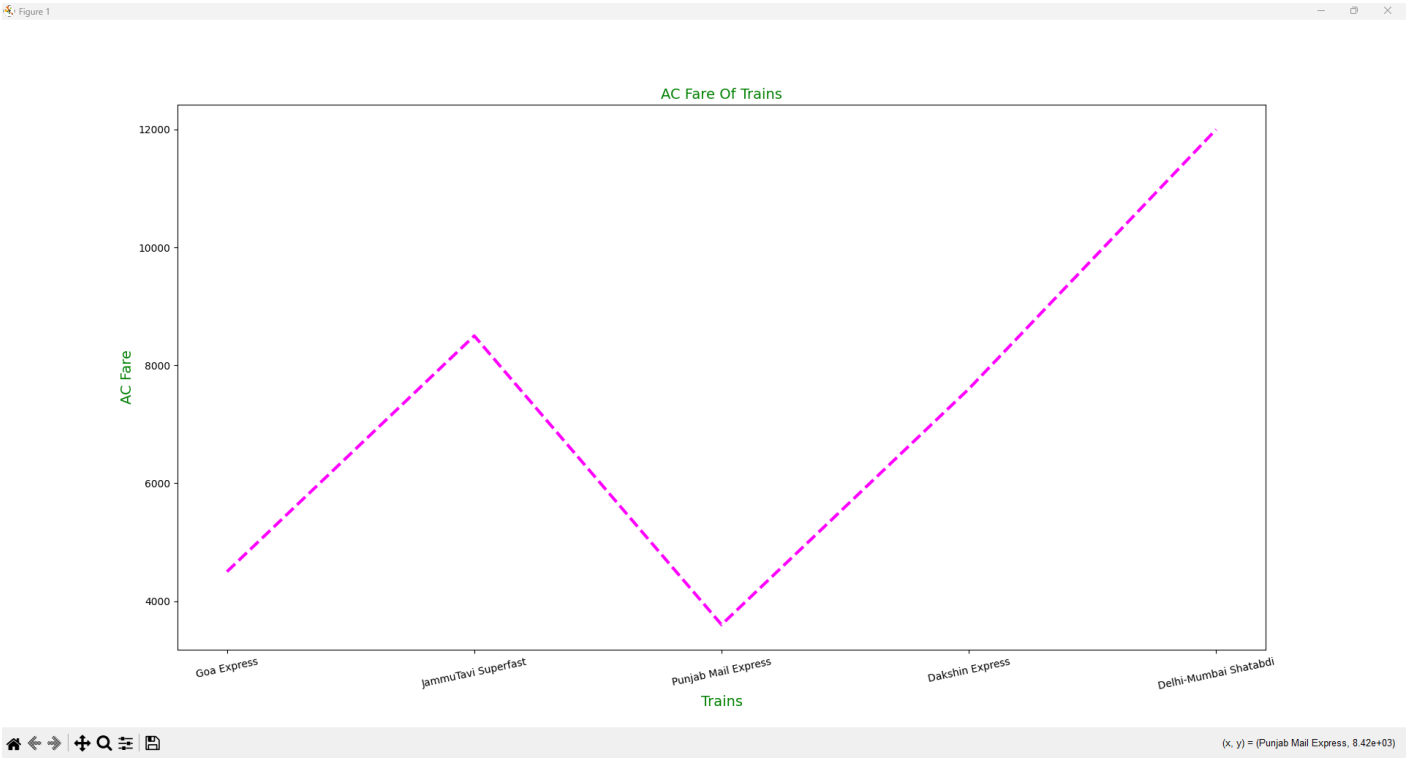
Analytics :

```
+=====================================================+
|                 ****  MAIN MENU  ****               |
+=====================================================+
|                                                     |
|  1. BAR PLOT :-   Fare  of  Passengers              |
|  2. LINE PLOT :-  AC Fare  Of  Trains               |
|  3. BAR PLOT :-   Sleeper Fare Of  Trains           |
|  4. LINE PLOT :-  Average Distance Covered By Train |
|                                                     |
+=====================================================+
```

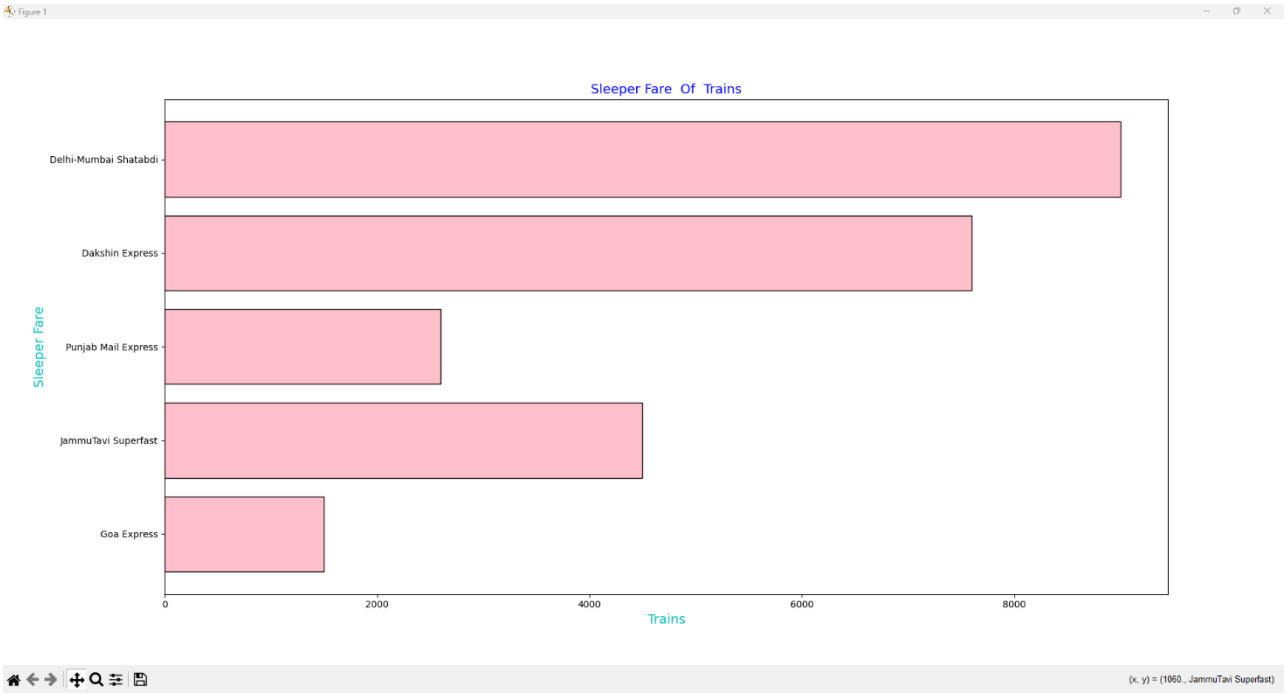
1.BAR PLOT :- Fare of Passengers :



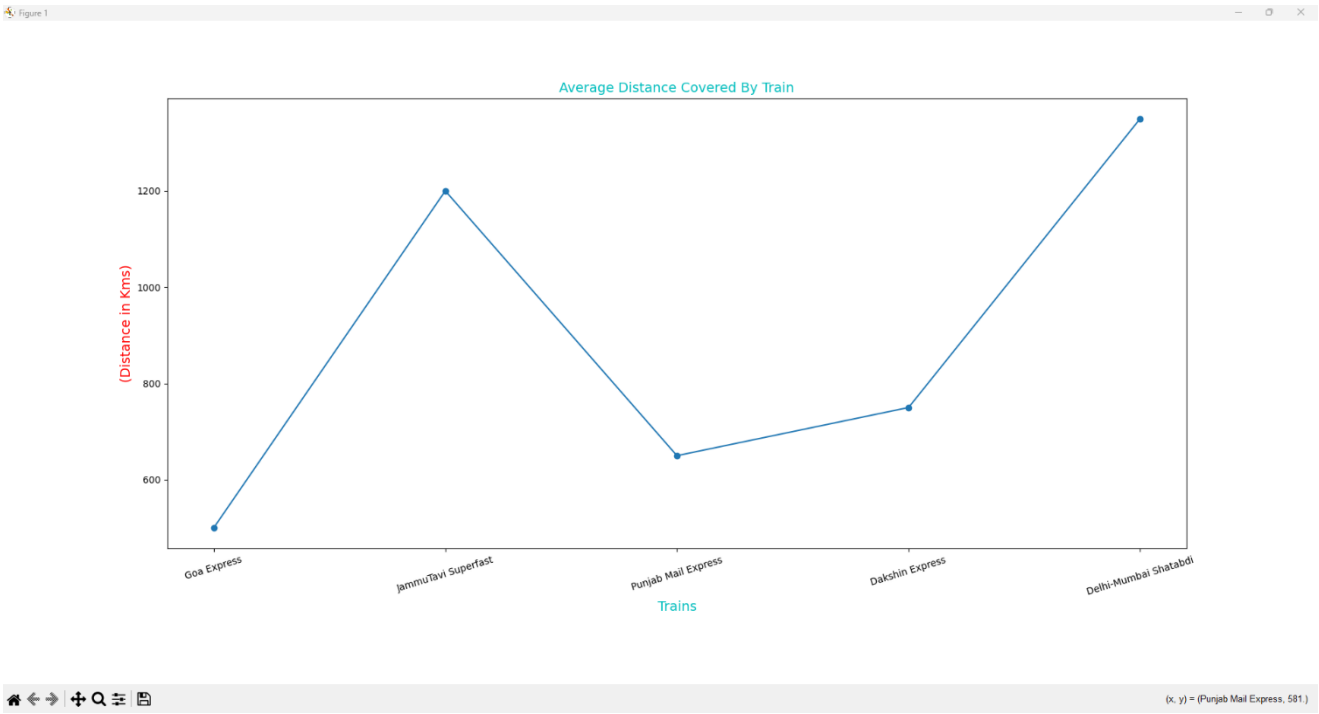
2.LINE PLOT :- AC FARE OF TRAINS :



3. BAR PLOT :-Sleeper Fare Of Trains :



4.LINE PLOT :- Average Distance Covered By Train :



EXIT :

```
##=========================================================##
||                        THANK YOU                        ||
##=========================================================##


##=========================================================##
||       DEVELOPED BY MOHD FAHAD XII-A 2024-25             ||
##=========================================================##
```

REFERENCES

1. WIKIPEDIA
   HTTPS://WWW.WIKIPEDIA.ORG/

2. PYTHON
   HTTPS://WWW.PYTHON.ORG/

3. MYSQL
   HTTPS://WWW.MYSQL.COM/

4. CLASS 11TH & 12TH COMPUTER SCIENCE SUMITA ARORA